

Problem Statement ID: 26158

Problem Statement: *Single
Pass Drone Video to Accurate
3D Model Reconstruction*

Theme: Robotics and Drones

PS Category: Software

Team ID: 162331

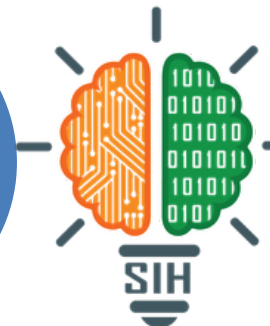
Team Name: Terraform





Single Pass 3D Mapping

FROM ONE FLIGHT → A 3D WORLD

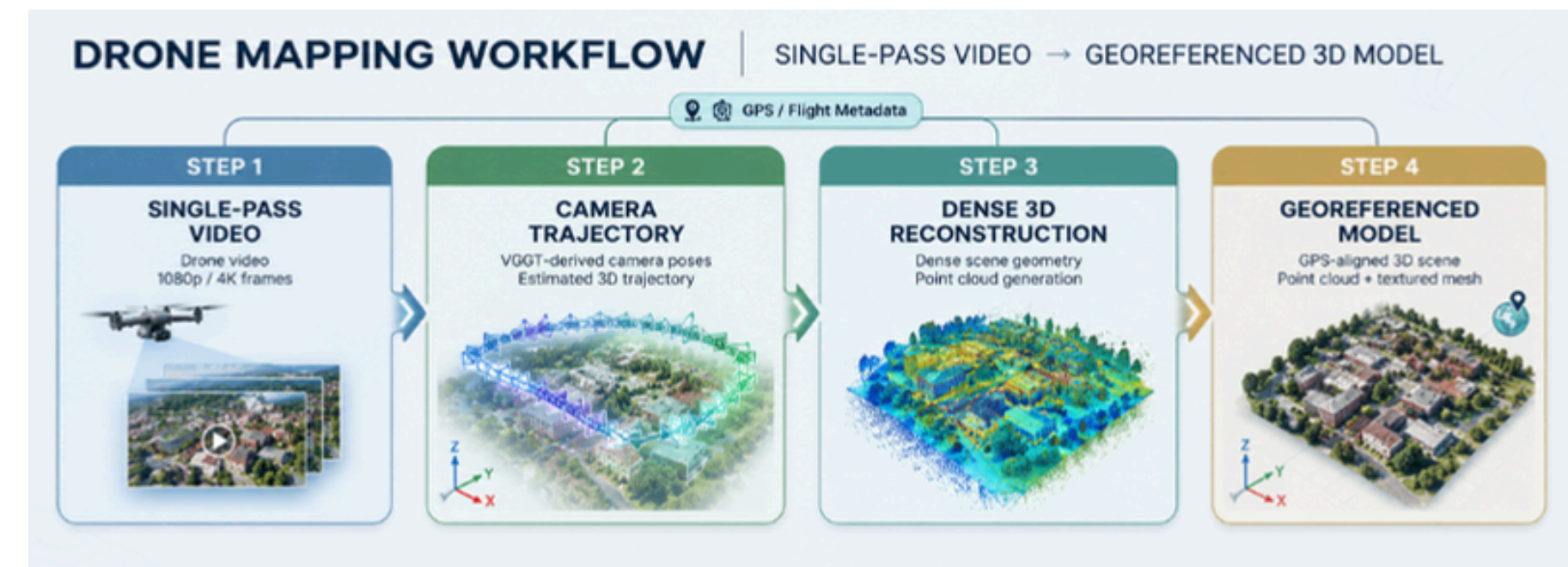


SMART INDIA
HACKATHON
2026

Objective

In disaster response and defence reconnaissance, only a single drone pass may be possible — which makes conventional multi-view photogrammetry, with its orbits and ground-control surveys, impractical. TERRAFORM reconstructs a metrically accurate, georeferenced 3D model of terrain, structures and infrastructure from one continuous pass, giving rapid situational awareness without a second flight.

Trajectory of Proposed Solution



Solution Proposed: Drone-Based Demo

01 | CAPTURE

Single-pass drone video + available flight/GPS data

02 | RECONSTRUCT

VGGT-based inference → camera poses + dense scene geometry

03 | GEOREFERENCE

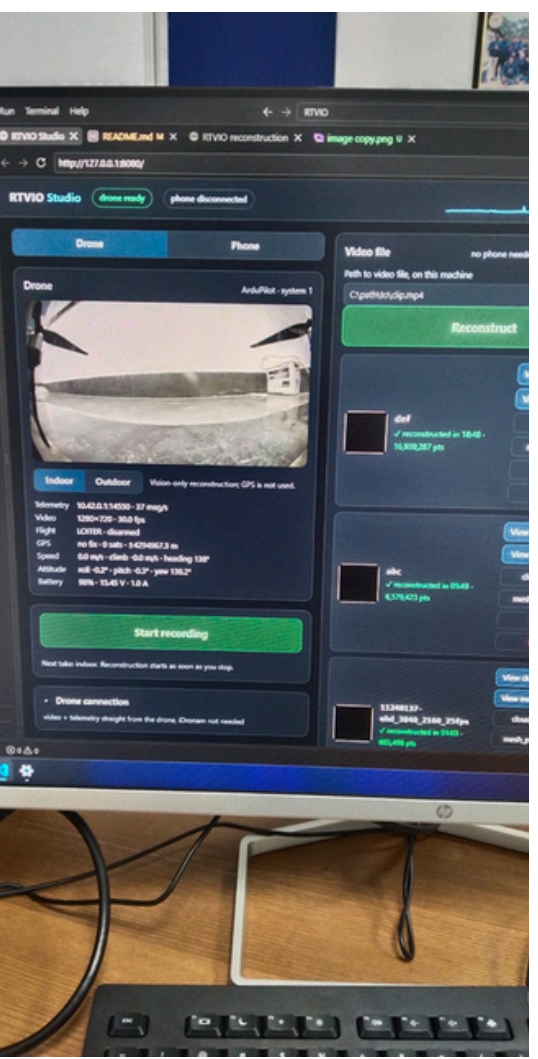
GPS-guided alignment → real-world coordinates

04 | DELIVER

Interactive 3D model + standard 3D/GIS exports

05 | VALIDATE

Per-run report: seam residuals, motion blur, point/mesh stats, timing



Strongest Part of Our Project

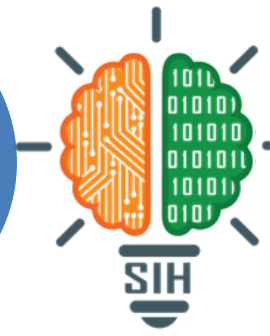
It already runs on real hardware, end to end.

- Real drone, ArduPilot telemetry + 1280×720 RTSP: 184 s indoor take (hand-carried), 5,520 frames, 0 dropped: recorded and reconstructed automatically.
- Real phone (Android 14), fully offline: 118.9 s, 2,863 frames, 0 dropped, 115 GPS fixes: transferred and reconstructed with no format conversion.
- Live path: a 101 s session produced a finished reconstruction 141.1 s after connecting, with most of the VGGT compute overlap
- Every run writes its own measured report. All hardware figures in this are labelled as such.



Single Pass 3D Mapping

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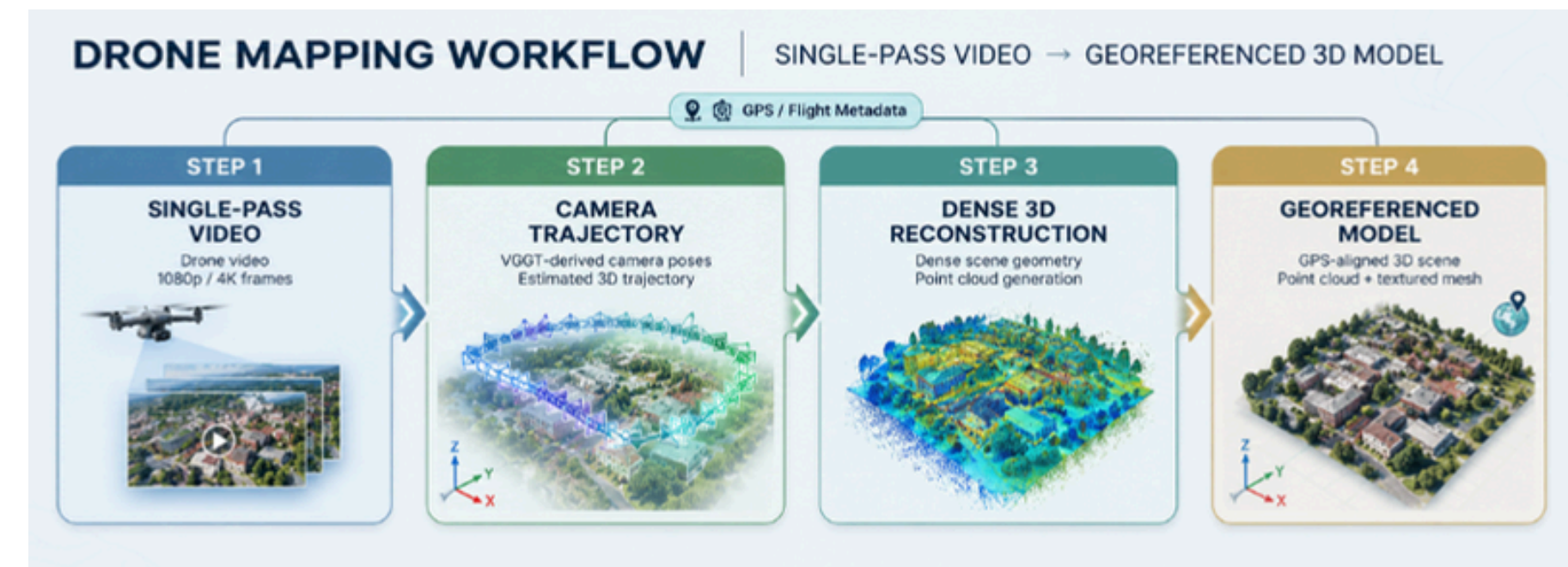


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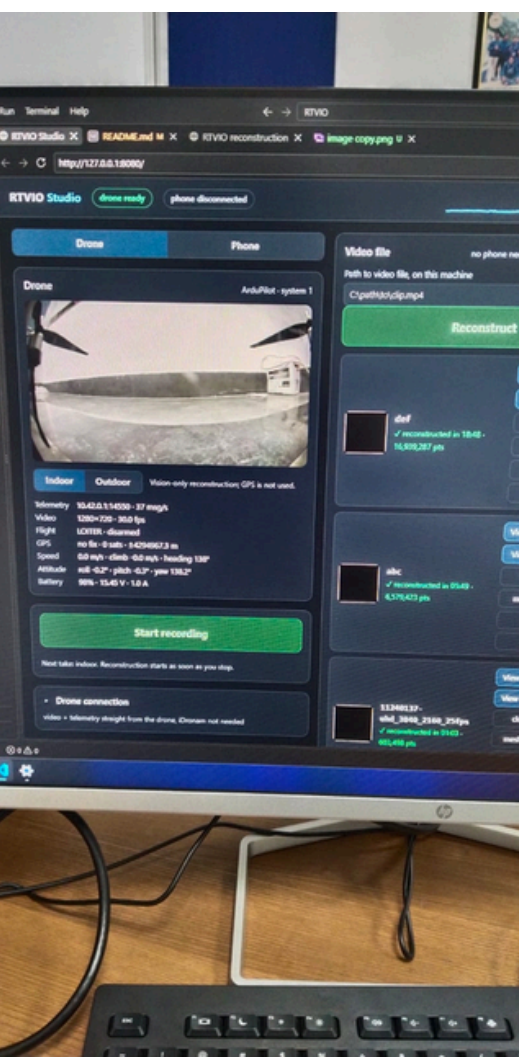
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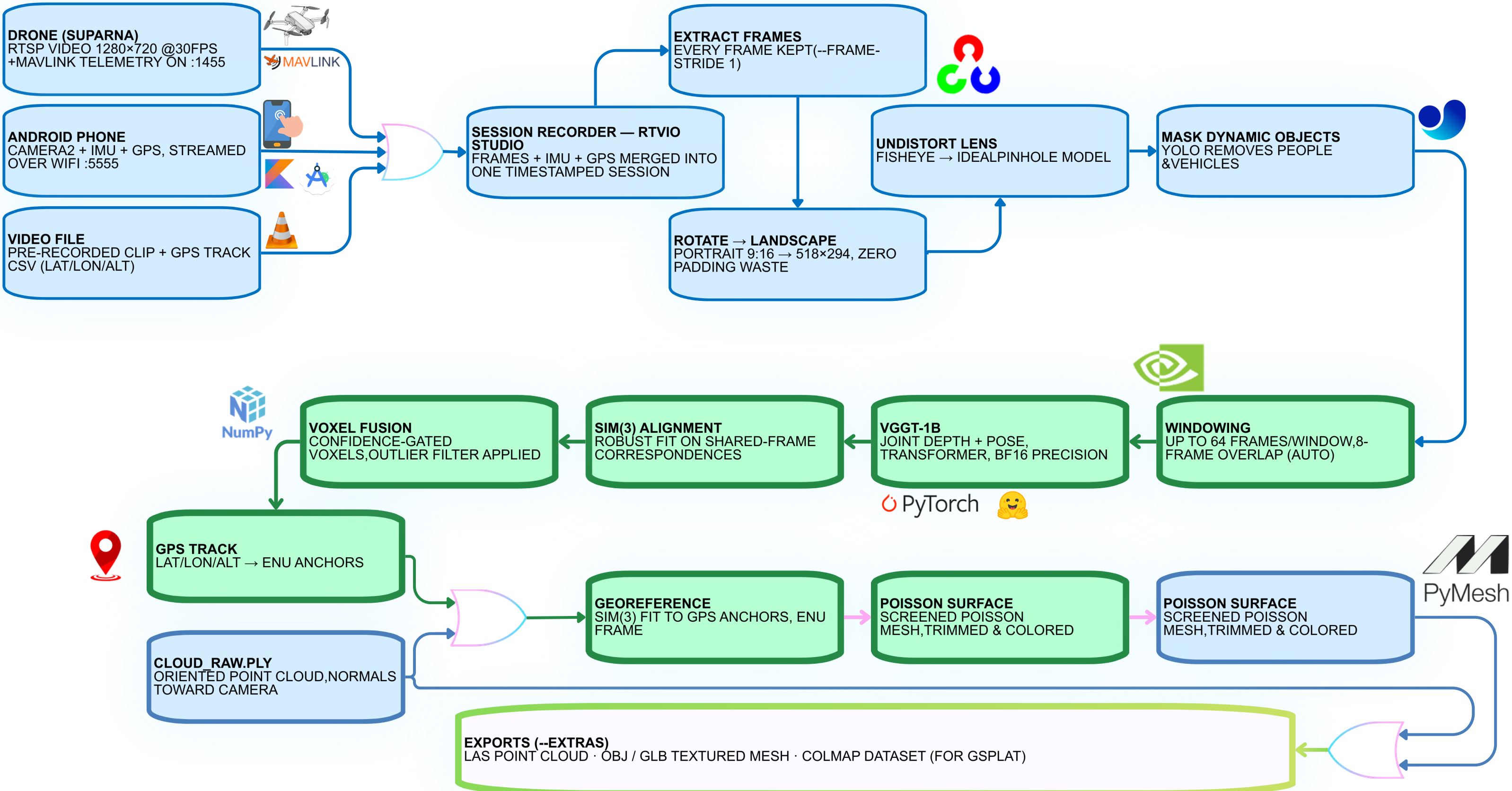
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- Live path: a 101 s session produced a finished reconstruction 141.1 s after connecting, with most of the VGGT compute overlapping the capture.
connecting : 0.72× real time, reconstruction overlapping capture.
- Every run writes its own measured report. Nothing in this deck is projected from a demo clip.





Technical Approach

Real-Time Visual-Inertial-Optical Reconstruction (RTVIO)

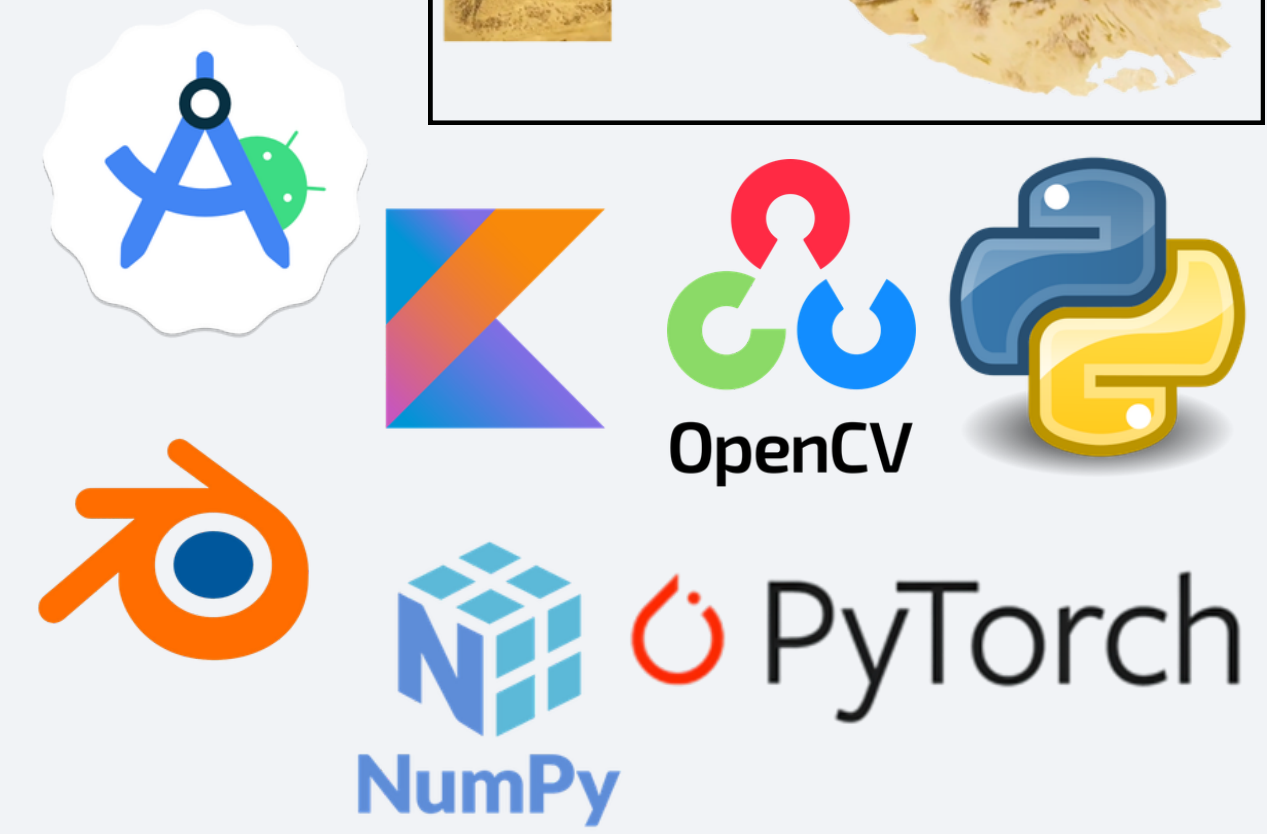
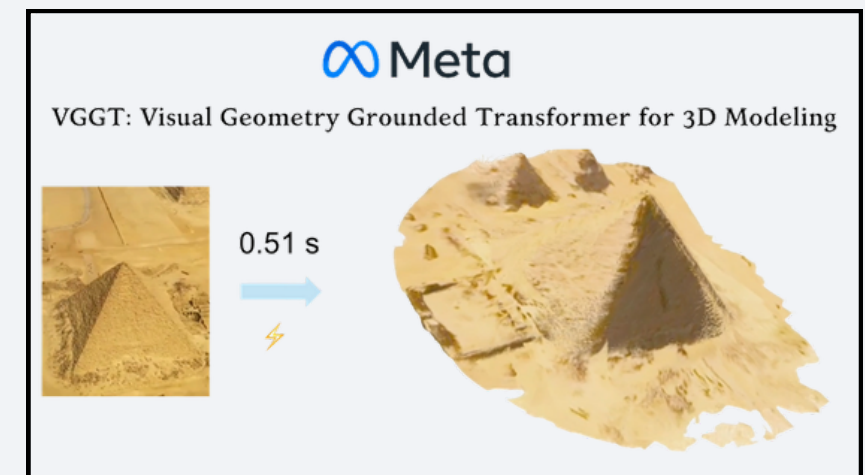
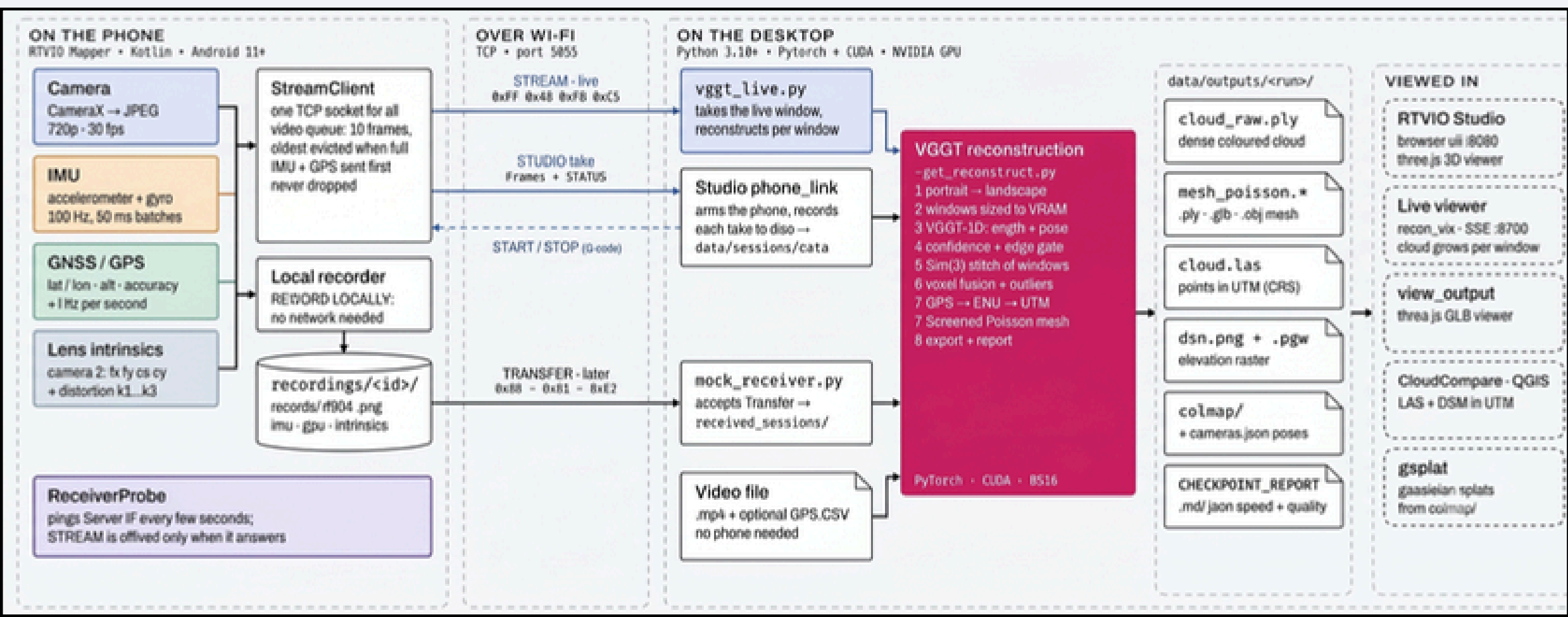


System Overview

Using ingestion checks and a pose-seeded MVS, the pipeline handles low-quality inputs like compressed video and blurred frames. Sharpness scoring + RANSAC-robust tracking absorb motion blur

Video, IMU and GPS leave a phone or drone over Wi-Fi. A GPU desktop turns the into a dense point cloud, a textured mesh and a UTM elevation raster, then shows the result in a browser.

Technologies Used



How it addresses the problem?

- Directly targets the single-pass constraint**
The pipeline is specifically designed for single-pass data, meaning no orbiting or multiple flights are required to reconstruct the scene.
- GPS-constrained similarity alignment**
By fitting a robust Umeyama Sim(3) transform (RANSAC-seeded, Cauchy-weighted) between GPS fixes and the reconstructed trajectory, the pipeline recovers a metrically scaled, georeferenced trajectory without needing dense Ground Control Point surveys.
- Sharpness scoring + RANSAC-robust tracking absorb motion blur**
Every frame gets a blur score in the run report, and consecutive VGGT windows are stitched with a RANSAC Sim(3) fit, so compressed or blurred frames don't break
- Pose-seeded MVS skips the costliest SfM**
This stage directly uses the recovered georeferenced trajectory to reconstruct terrain and facades, targeting near-real-time results.



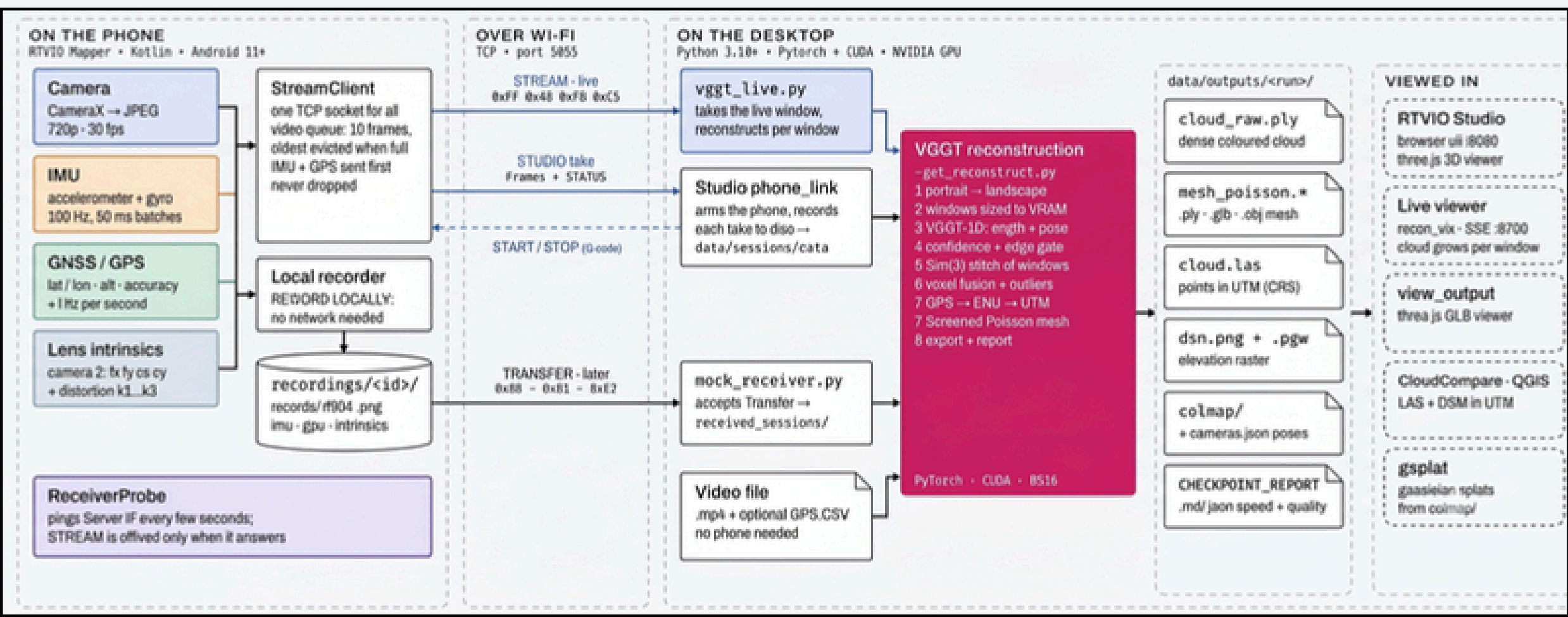
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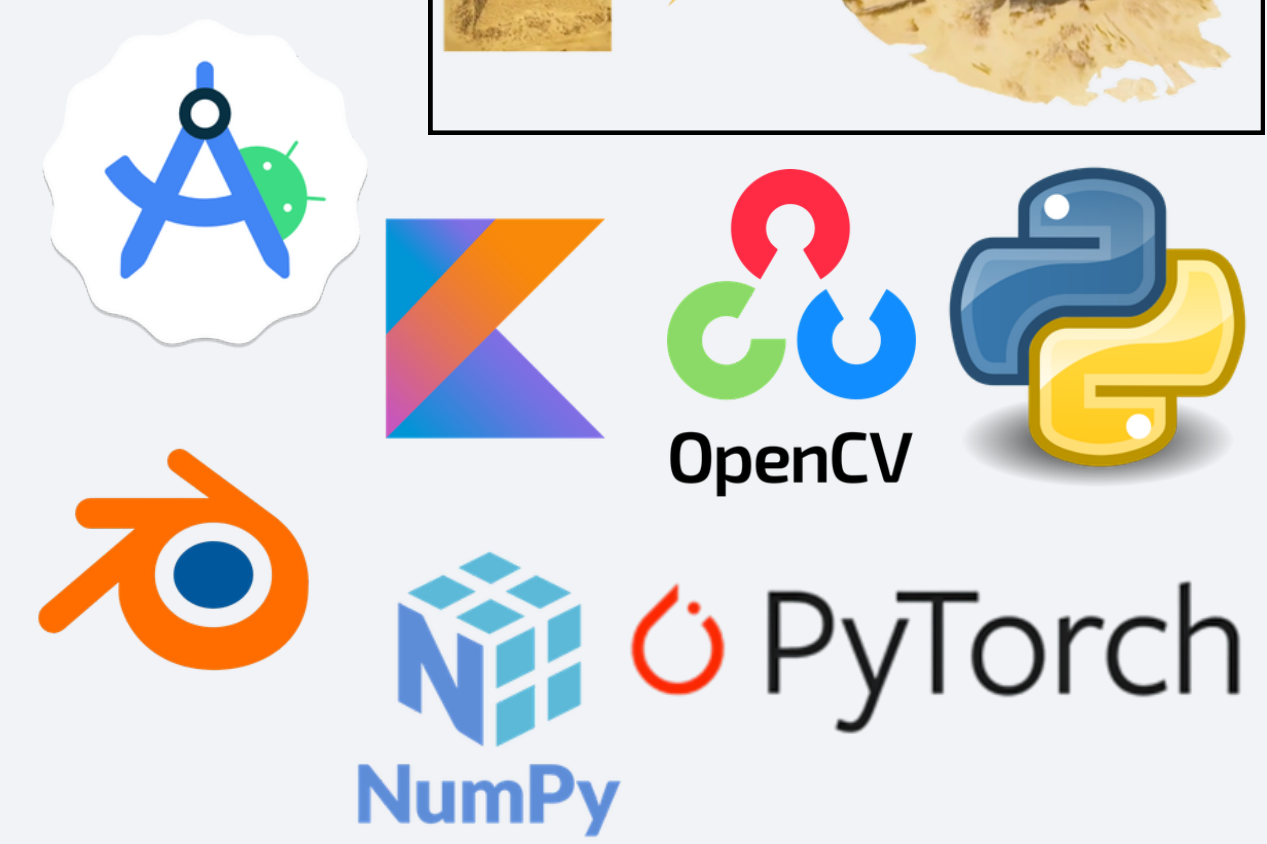
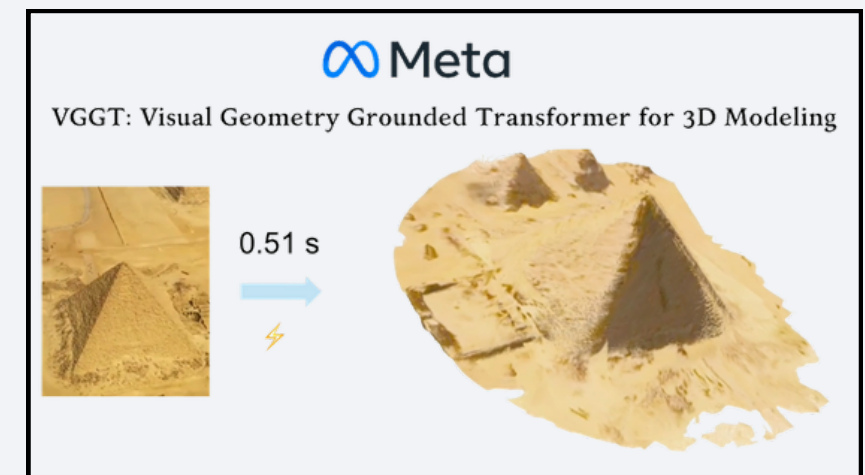


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- VGGT replaces the costliest SfM + MVS stages**
One transformer pass predicts camera poses and dense depth together, with no COLMAP matching, so terrain and facades are reconstructed window by window at near-real-time speed.



Feasibility and Vialibility



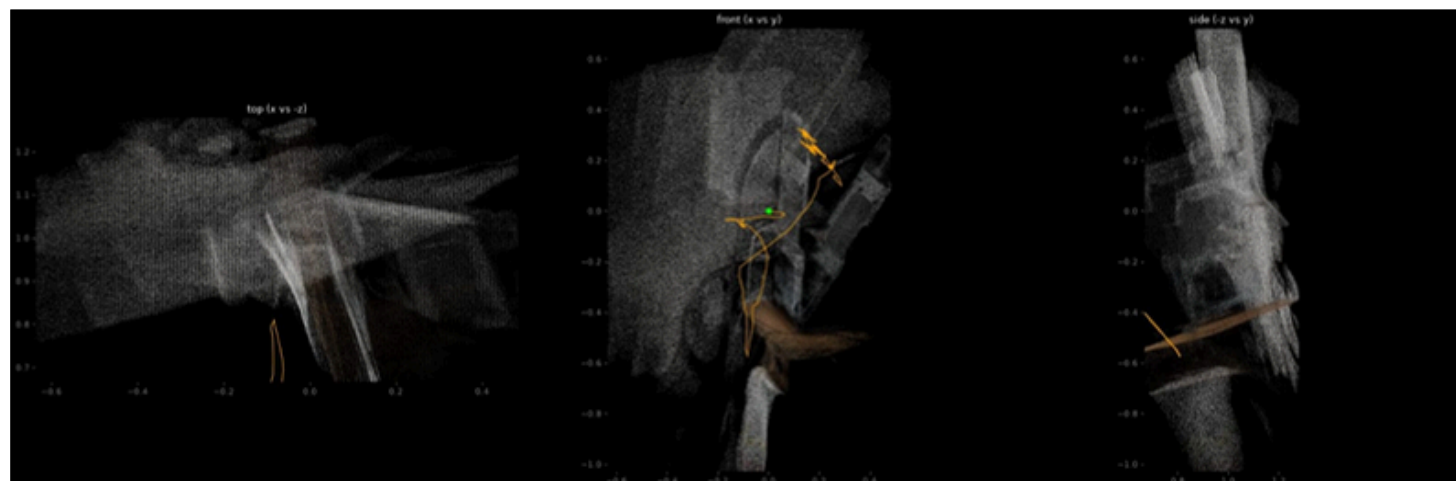
TECHNICAL FEASIBILITY

- Core pipeline works end-to-end VGGT reconstruction, Poisson meshing, and export (OBJ/PLY/LAS/gITF) are already running on real drone footage with measured output, not theoretical.
- Georeferencing has a closed-form solution : Umeyama alignment (SVD-based) guarantees an optimal rigid+scale transform, with error shrinking as $\sigma/\sqrt{N}$, making $\leq 1\text{m}$ accuracy achievable even under GPS noise.
- Speed gap is known and tunable : measured per-window cost ($\sim 22.7\text{s}$) shows the current setup exceeds the 15-min budget on long videos, but scales predictably with $\text{SAMPLE_FPS}/\text{WINDOW_FRAMES}$, giving a clear mitigation path rather than an unknown risk.

PROOF OF IMPLEMENTATION

Real 3D reconstruction generated by our pipeline from captured imagery

Input: PC/CPU scene imagery
Output: 3D point cloud + camera trajectory



POTENTIAL CHALLENGES

LONG VIDEO RUNTIME

- Challenge:** Current benchmark projects $\sim 15\text{min}$ for a 10-min video.
- Mitigation:** Optimize frame sampling, window size and inference speed.

GPS NOISE

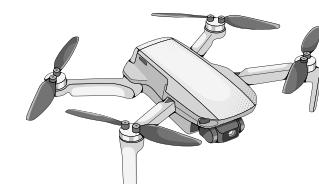
- Challenge:** Consumer GPS introduces meter-level errors.
- Mitigation:** Test 1 m / 3 m / 5 m noise and quantify alignment error and use RTK Sensors

DYNAMIC OBJECTS

- Challenge:** Cars and moving objects can corrupt reconstruction.
- Mitigation:** Evaluate aerial-trained segmentation.

OCCLUDED SURFACES

- Challenge:** A single flight cannot observe every surface.
- Mitigation:** Flag low-confidence/unobserved regions instead of hallucinating geometry.



Economic Impact

Reduces cost and time of land surveys, construction monitoring, and infrastructure audits — valuable for governments and private sectors managing large-scale assets.

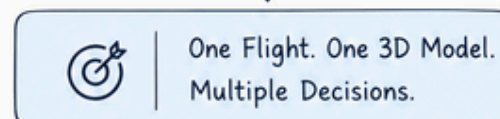
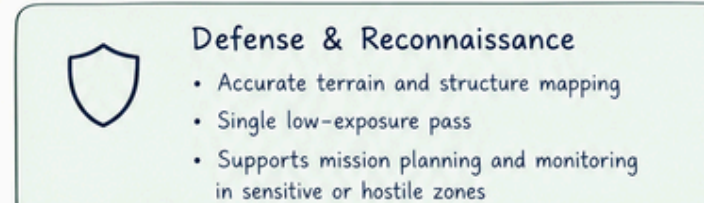
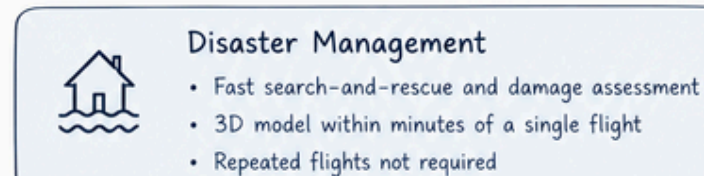
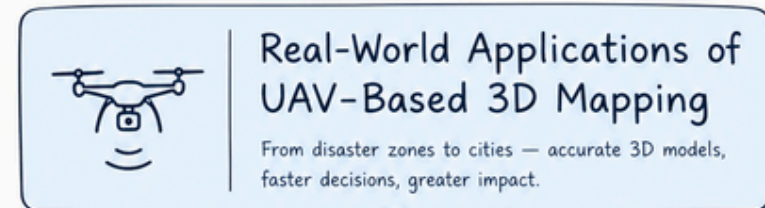
Agricultural Impact

The same single-pass approach can extend to crop/field monitoring - terrain and vegetation mapping for precision agriculture, at low cost.

Digital Twin Creation for Smart Cities

Delivers rapid, low-cost 3D city models that can be frequently updated for planning, traffic simulation, and infrastructure management. The open, reproducible pipeline enables transparent, data-driven urban governance without recurring license fees.

Social and Administrative Benefits



Government-Ready Accountability: Open Audits and Clear Usage Policy

Open, government-auditable pipeline

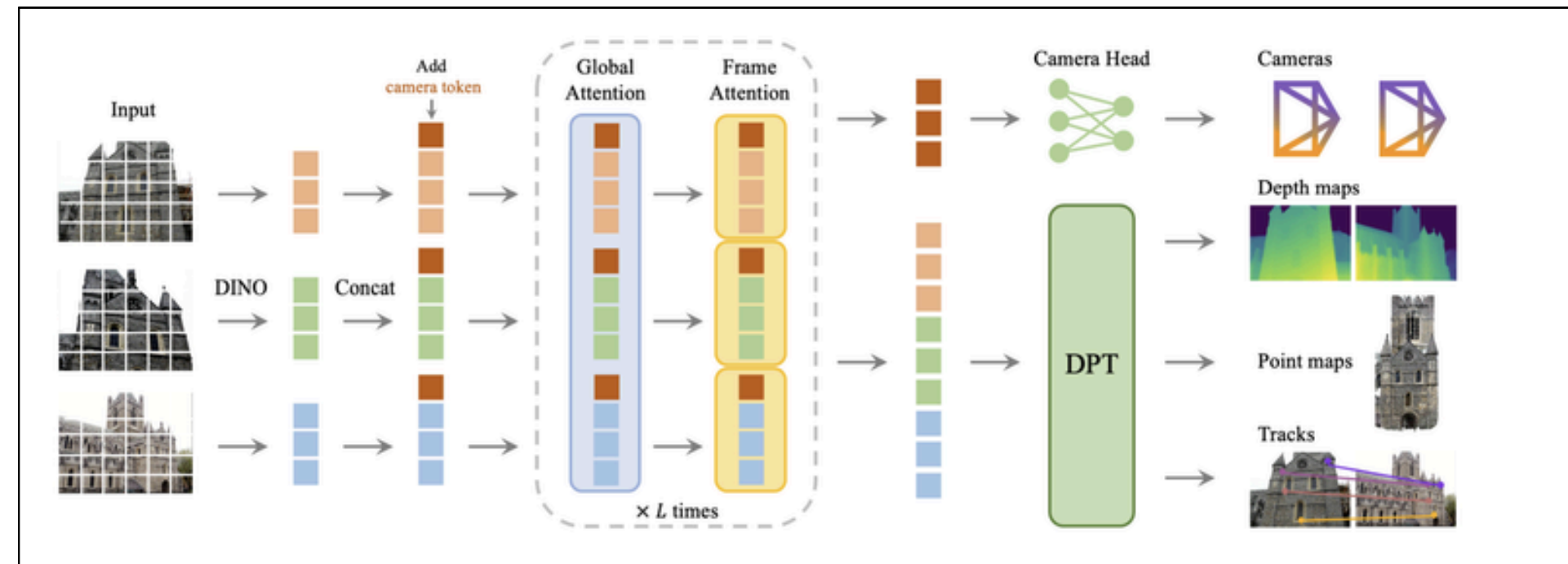
- All core code and benchmarks (UseGeo, synthetic data) are free and public, so any ministry, audit agency, or state lab can independently verify accuracy, speed, and limitations without buying proprietary software.
- Enables transparent procurement and evaluation: governments can compare this against commercial tools (Pix4D/Agisoft) on the same public datasets and published metrics.

Clear Policy-Aligned Usage Guidance

- Provide simple, official-style guidance on where outputs are suitable (rapid disaster assessment, preliminary planning, reconnaissance) and where they are not (legal land records, structural safety certification) without ground control points.
- Helps governments adopt the system responsibly in disaster management, defense, and urban planning while managing liability and public trust.

SUMMARY: Our approach is grounded in established research on multi-view geometry, neural 3D reconstruction, and geometric alignment. VGGT provides the core visual reconstruction, while SfM and Multiple View Geometry establish the underlying principles. Umeyama alignment enables georeferencing, and the UseGeo UAV–LiDAR dataset provides a basis for quantitative validation.

Research → Implementation → Validation



VGGT - Visual Geometry Grounded Transformer, Wang et al., CVPR 2025

[01 - VGGT: Visual Geometry Grounded Transformer J. Wang et al. - CVPR 2025](#)

Core 3D reconstruction model used in TERRAFORM. Predicts camera parameters, depth maps, point maps and 3D point tracks from multiple views.

[02 - Structure-from-Motion Revisited J. L. Schönberger & J.-M. Frahm -CVPR 2016](#)

Foundational reference for multi-view reconstruction, camera estimation and sparse 3D structure.

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Mathematical foundation for camera projection, epipolar geometry, triangulation and 3D reconstruction.

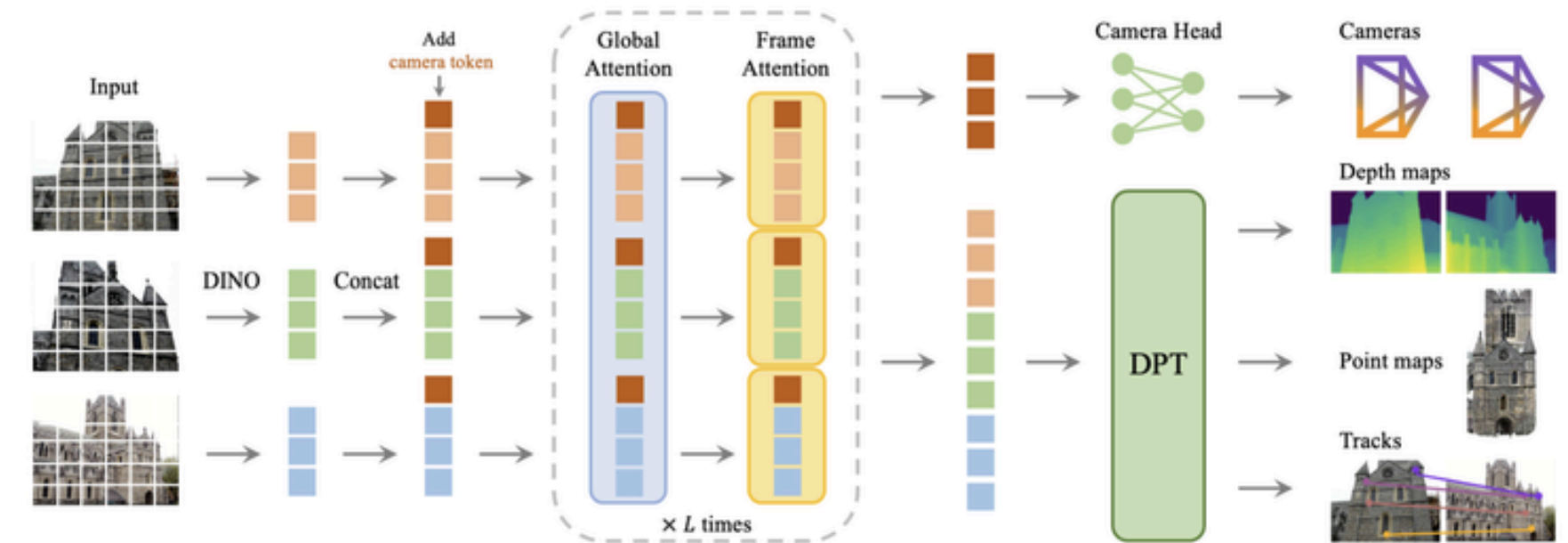
[04 - Least-Squares Estimation of Transformation Parameters Between Two Point Patterns, S. Umeyama - IEEE TPAMI, 1991](#)

Mathematical basis for the similarity transformation used in your GPS-based georeferencing: rotation, translation and scale.

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